MIL-L-52039D
29 November 1976
SUPERSEDING
MIL-L-52039C(ME)
19 November 1971

MILITARY SPECIFICATION

LIGHT, GLIDE-ANGLE, AIRPORT APPROACH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an airport approach glide-angle light designed to project three separate beams of light, one red, one green, and one yellow.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

FSC 6210

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra-Flexible, 300 and 600 Volts).
MIL-C-52286	- Cable Assemblies, Power, Electrical, (With Molded on Terminations).

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 101	- Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

TA13205E3560	- Light, Glide, Angle Airport Approach.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The glide-angle light shall be in accordance with Top Assembly TA13205E3560 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the airport approach glide-angle light. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more glide-angle lights as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior

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to starting production that his production methods will produce glide-angle lights that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Level-vial liquid. The level-vial liquid shall not freeze at minus 50° F and the bubble shall not expand beyond the marks on the vial at plus 100° F.

3.4 Cable assembly. The cable assembly shall consist of a 2-conductor cable conforming to MIL-C-3432, Type CO-02HLF (2/14) 0550 terminating in a male 2-pole connector plug conforming to MIL-C-52286, Figure 24. The cable assembly shall be 36 inches long.

3.5 Lamp. When specified, the glide-angle light shall be furnished with a lamp. The lamp shall be Industry No. 6.6/TIO/3P for use on 6.6 amperes series circuit or Industry No. 100T10 for use on 120-volt, single phase, ac circuit, as specified.

3.6 Operation. The glide-angle light shall operate without malfunction from a power source of 120 volts ac or from a 6.6 amperes series airport lighting circuit.

3.6.1 Leveling. The bubbles on the levels located on the protractor pointer and housing shall be centered when the protractor pointer is locked at zero.

3.6.2 Protractor. The protractor shall be mounted and alined so that when the pointer is set at zero reading, the center of the green beam and the pointer are not displaced more than plus or minus 15 minutes of arc.

3.6.3 Sights. The line of sight established by the front and rear sights shall be parallel to within 6 minutes of arc of the center of the green beam.

3.6.4 Light tightness. With the lamp in the glide-angle light operating at rated voltage or current, no light shall escape except through the color filters.

3.6.5 Dielectric strength. The elctrical circuitry shall withstand 600 volts, ac, 60 Hertz between each pin of power cable and 1000 volts ac, 60 Hertz between each pin and metal case of light without damage or arc over.

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3.7 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.7.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.7.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.8 Identification markings. The identification markings shall be in accordance with MIL-STD-130 and as specified on the drawings.

3.9 Workmanship. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety. Joints and seams shall be tight fit; edges shall be smooth, and optical surfaces shall be clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the

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contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any examination or tests by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6).

4.1.3 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The glide-angle light shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The glide-angle light shall be tested as specified in 4.5.2 through 4.5.2.6. Failure of any test shall be cause for performing the inspection specified in 4.1.1.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105 Inspection Level I.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for major and minor defects specified in 4.5.1. AQL shall be 6.5 percent defective for major defects and 10 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each glide-angle light shall be tested as specified in 4.5.2.2. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.3, 4.5.2.4, 4.5.2.5, and 4.5.2.6. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The glide-angle light shall be examined as specified herein for the following defects:

Major

101. Nonconformance to dimensions shown on the drawings.
102. Parts or components missing.
103. Misalinement of parts of components.
104. Materials or components not as specified.
105. Any part (or component) not in accordance with the SQAP requirements as shown on the drawings.
106. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification marking not as specified, or not as shown on the drawings.
203. Instruction plates not as shown on the drawings.
204. Any dimension other than those identified on the SQAP requirements shown on the drawings not as specified.

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4.5.2 Tests.

4.5.2.1 Level-vial liquid. A level vial containing the test liquid shall be placed in a cold chamber having an ambient temperature of minus 50° F at the end of 24 hours, the liquid in the vial shall be examined for evidence of freezing. The vial shall then be placed in a room or chamber with an ambient temperature of 100° F. At the end of 3 hours, the liquid in the vial shall be examined for evidence of expansion beyond the limiting marks on the vial. Freezing of the liquid or expansion beyond the limiting marks shall constitute failure of this test.

4.5.2.2 Operation. The glide-angle light, with an incandescent lamp installed, shall be connected to either a 120-volt ac power source or a 6.6 ampere series lighting circuit and operated. The incandescent lamp used in this test shall be of the type specified for the power supply used. The contractor shall furnish the lamp. Inability to operate, short circuits, open circuits, or arcing shall constitute failure of this test.

4.5.2.3 Leveling. The glide-angle light shall be mounted in its normal operating position and one spirit level shall be placed transversely and one longitudinally on the top of the glide-angle light with the protractor pointer locked at zero on the scale. The test-level bubbles shall be centered. Determine whether the installed levels on the pointer and back cover of the glide-angle light are centered. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.4 Protractor. The lamp in the glide-angle light shall be turned on, and with the protractor pointer set at zero, the level-vial bubble shall be centered and the center of the projected green light shall be marked on a vertical surface. The protractor arm shall then be reset at 15 degrees of arc on the scale and locked, the arm leveled, and the center of the projected green light marked on the surface. The angle between both marks shall be measured. A difference of more than plus or minus 15 minutes of arc between the measured angle and 15 degrees of arc shall constitute failure of this test.

4.5.2.5 Sighting. The light in the glide-angle light shall be turned on and the light beam shall be projected on a vertical surface 10 feet from the front of the light. The center of the green beam shall be marked on the surface. A sight shall be taken between the bottom edge of the rear rectangular sight and the bottom edge of the front rectangular sight, an extension of the sight line shall be marked on the surface. The vertical surface shall be moved to a position 20 feet from the front

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of the light and the above procedure shall be repeated. A difference of more than 0.2 inch in the separation between the two marked lines at 10 feet and a difference of more than 0.4 inch in the separation between the two marked lines at 20 feet shall constitute failure of this test.

4.5.2.6 Light tightness. The glide-angle light, with the filter masked to prevent light leaks, shall be placed in a darkroom or other area having an ambient illumination not greater than 10-5 foot-candles and shall be operated in accordance with 4.5.2.2. Three observers, with vision corrected to 20/20 and dark adapted for a minimum of 30 minutes under illumination not greater than 10-5 foot-candles (darkroom with no visible light), shall observe the glide-angle light from all angles from a distance of 25 feet. Detection of white light by two observers shall constitute failure of this test.

4.5.2.7 Dielectric withstanding voltage test. With lamp removed from lamp receptacle, dielectric strength test shall be performed in accordance with Method 301 of MIL-STD-202. Using test voltage of 600 ac, 60 Hertz for test between each insulated pin in power cable and using test voltage of 1000 ac, 60 Hertz for test between pins in power cable and metal case of glide angle light. Nonconformance to 3.6.5 constitutes failure of test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The preproduction pack of the lights for Level A or B shall be subjected to the free-fall drop test in accordance with FED. TEST METHOD STD. No. 101, Test Method 5007, Procedure E, and Procedure B. At the conclusion of the preproduction pack test, the pack and all contained packages shall be opened and examined along with their contents. Rupture of the pack or any package or any visible damage to the contents shall constitute failure of the preproduction pack and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105 Inspection Level S-2.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following major and minor defects. AQL shall be 6.5 percent defective for major defects and 10.0 percent defective for minor defects.

Major

107. Material and container not as specified. Each incorrect material or container shall be considered one defect.
108. Light not secured within the box as specified for Level A.
109. Box not waterproofed as specified for Level A or B.
110. Gross weight of boxes exceed 200 pounds for Level A or B.
111. Strapping not as specified for Level A.

Minor

205. Fiberboard boxes exceed the size and weight limitations for Level B.
206. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and tests within the time frame specified, to prove prior to starting production packaging that the applied preservation packing and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction model glide-angle or a production glide-angle light. If the preproduction model glide-angle light is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model glide-angle light and production glide-angle light.

5.2 Preservation. Preservation shall be Level A or Commercial as specified.

5.2.1 Level A. Each light with technical publications shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. The light shall be secured within the box in a manner to prevent movement

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and damage to the light by use of fiberboard open-end cells or corner pads. Fiberboard used for the cells or pads shall conform to PPP-F-320, Type CF, Class Weather-Resistant, Variety SW, Grade W6c. The box shall be closed and waterproofed as specified for slotted style boxes in accordance with the appendix to the box specification.

5.2.2 Commercial. Lights, with technical publications, shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.3 Packing. Packing shall be Level A, B, or Commercial, as specified.

5.3.1 Level A. Lights shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, Grade B. The gross weight of each box shall not exceed 200 pounds. Strapping shall conform to QQ-S-781, Class I or VI, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.3.2 Level B. Lights shall be packed in close-fitting boxes conforming to PPP-B-601, Domestic Type, Grade B, PPP-B-640, Class 2, style optional, or PPP-B-636, V3c, style optional. The gross weight of each box shall not exceed the limitations of the box specification or 200 pounds, whichever is less. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Fiberboard boxes shall be waterproof sealed.

5.3.3 Commercial. Lights shall be packed to assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Commercial packaging marking shall reflect as a minimum, the item name, National Stock Number, quantity, contractor purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible and shall provide a definite contrast.

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6. NOTES

6.1 Intended use. The glide-angle lights are intended for use to guide a pilot approaching a landing field to the proper glide angle for descent to the touchdown area.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of preproduction models required (see 3.2).
(c) Whether lamp is required and, if so, type of lamp required (see 3.5).
(d) Degree of preservation and packing required (see 5.2 and 5.3).
(e) When Finish A strapping is required (see 5.3.1).

6.3 Preproduction model. Any changes or deviations of production glide-angle lights from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish glide-angle lights conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the glide-angle lights in accordance with this specification.

6.5 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of the classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

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6.6 Definitions.

6.6.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

Custodians:
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Review activity:
DSA - GS

User activity:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-L-52039D Light, Glide-Angle, Airport Approach

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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